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AIM:

To design and configure IPv6 addressing for a network of 8 systems using Cisco packet Tracer and verify communication between the systems using ping command.

Procedure:

- open Cisco packet tracer.
- Place 8 pcs and one 2960 switch in the workspace.
- Connect all the pcs to the switch using copper straight-through cables.
- Configure IPv6 addresses for all PCs using the network 2001:DB8:1::/64.
- Assign unique IPv6 addresses to each PC.
- open Desktop → IP Configuration on each pc and select IPv6 static.
- Enter the respective IPv6 address and prefix length 64.
- open Desktop → Command prompt on a pc.
- use the ping command followed by destination IPv6 address.
- verify successful communication by checking the reply message.

~~IPv6~~ Addressing.

Sample output:

Ping 2001:DB8:1::8

Pinging 2001:DB8:1::8 with 32 bytes
of data.

Reply from 2001:DB8:1::8

Reply from 2001:DB8:1::8

Reply from 2001:DB8:1::8

Reply from 2001:DB8:1::8

Ping successful.

Result:

The IPv6 addressing for 8 systems
was successfully configured using Cisco
packet Tracer, and communication between
the systems was successfully verified using
the IPv6 ping command.

2, AIM:

To configure a basic site to site VPN tunnel between the head office and Branch office of a bank using Cisco packet Tracer and verify secure communication between the two networks

Procedure:

1. Open Cisco packet Tracer and Create the following topology:

- 2 Routers
- 2 Switches
- 2 PCs
- Connect them

2. Configure the IP addresses

3. Configure the VPN on both routers using ISAKMP policy, Pre shared Key, IPsec transform set, access list and Crypto map.

4. From PC₁, test communication with PC₂
Ping 10.2.2.10

5. Verify the VPN using.

sample output.

Ping 10.2.2.10

Reply from 10.2.2.10: bytes = 32

Reply from 10.2.2.10: bytes = 32

Reply from 10.2.2.10: bytes = 32.

Reply from 10.2.2.10: bytes = 32

Result:

Site to site IPsec VPN tunnel was successfully configured between the Bank head office

and Branch office using Cisco packet
Tracer and communication between
the two networks was successfully verified

3, AIM:

To capture and analyze RTP/VOIP packets using Wireshark and study their header fields.

Procedure:

1. Open Wireshark and select the active network interface.

2. Start packet capture.

3. Generate / observe a VOIP call.

4. Stop the capture.

5. Apply the filter.

rtp

6. Select an RTP packet and expand Real time Transport protocol.

Observe the RTP header fields such as Version, payload Type, sequence Number, Timestamp and SSRC.

Sample output:

RTP header.

Version : 2

Payload Type : PCMU (0)

Sequence number : 1253

Timestamp : 201328640

SSRC : 0x3A7B21C4

Result.

Thus, the RTP, VOIP packets were successfully captured and analyzed using Wireshark, and their header fields were identified.

4, AIM:

To design a point to point topology connecting two routers through a serial link and configure IP addressing on both ends.

Procedure:

- Open Cisco packet Tracer.
- Place 2 routers (R₁ and R₂) in the workspace.
- Connect R₁ and R₂ using a serial DCE Cable
- Configure IP addresses on the serial interfaces.
- Verify the connection using:
show ip interface brief.
- Test connectivity from R₁
Ping 192.168.1.2

Sample output:

Success rate is 100 percent (5/5)

Result:

Thus, a point to point topology was successfully created and IP addressing was configured on both routers using a serial link.



